



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
TKM COLLEGE OF ENGINEERING, KOLLAM

OPTICAL CHARACTER RECOGNITION FOR ENGLISH LANGUAGE

PHASE II PROJECT EVALUATION

TEAM 14

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DATE : 08/04/25

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1. Introduction
2. Motivation
3. Literature Review
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INTRODUCTION

- Optical Character Recognition
 - Scanned documents/Images -> editable digital data.
- Challenges:
 - Noisy images
 - Complex layouts
 - Handwriting

MOTIVATION

- Integrate to digital workflow
- Diverse document types
- Efficiency in accessibility
- Digital India Initiative

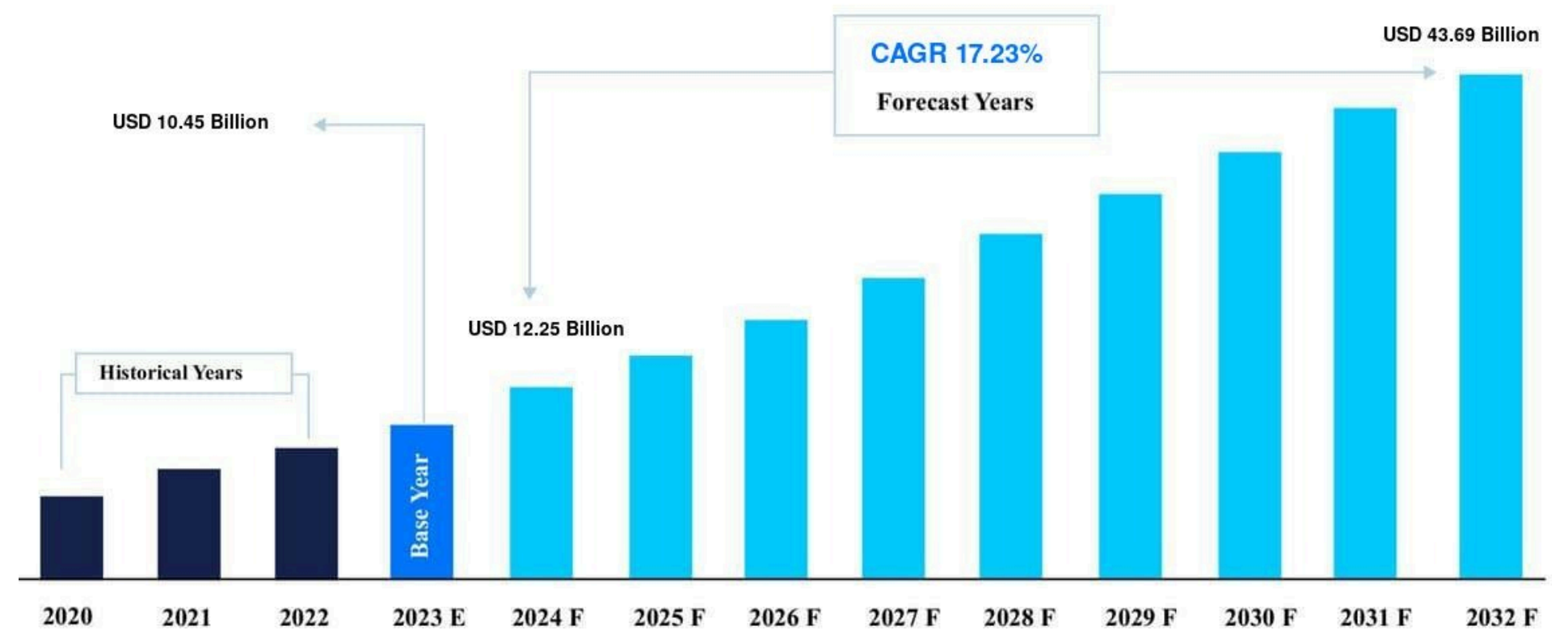


FIG 1: DIGITAL INDIA FORECAST

LITERATURE REVIEW

TITLE	TECHNOLOGIES USED	INFERENCE
TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models (M. Li et al., 2021)	TrOCR	Delves into transformer based models and its advantages over existing models
PaddleOCR: An Ultra-lightweight OCR Toolkit Based on PaddlePaddle (Baidu, 2020)	PaddleOCR	Delves into PaddleOCR architecture and its relevance in handwritten text recognition
Which OCR toolset is good and why? A comparative study (Pulkit Batra et al., 2023)	TesseractOCR, KerasOCR, PaddleOCR, Microsot Azure Computer Vision	Comparative study that highlights the strengths and weaknesses of differernt OCR methodologies

ARCHITECTURE

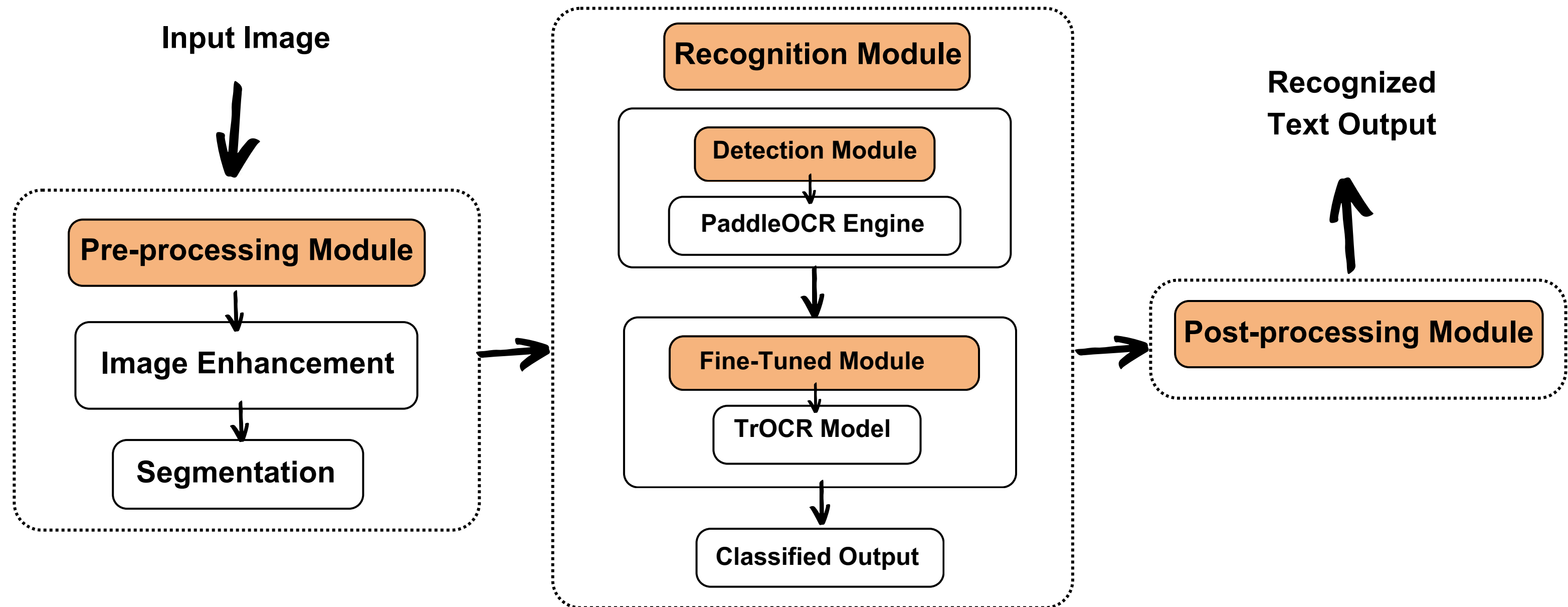


FIG 2: ARCHITECTURAL DESIGN

PROPOSED PHASES

METHODOLOGY

1. Data Collection
2. Preprocessing Module
3. Recognition Module
4. Training and Validation
5. Postprocessing Module

1. DATA COLLECTION

METHODOLOGY

- IAM Form Dataset:
 - Diverse handwriting styles:
 - Slant
 - Spacing
 - Pen pressure
 - Single unit data:
 - Preprocessing
 - Line/word segmentation
 - Annotation file creation

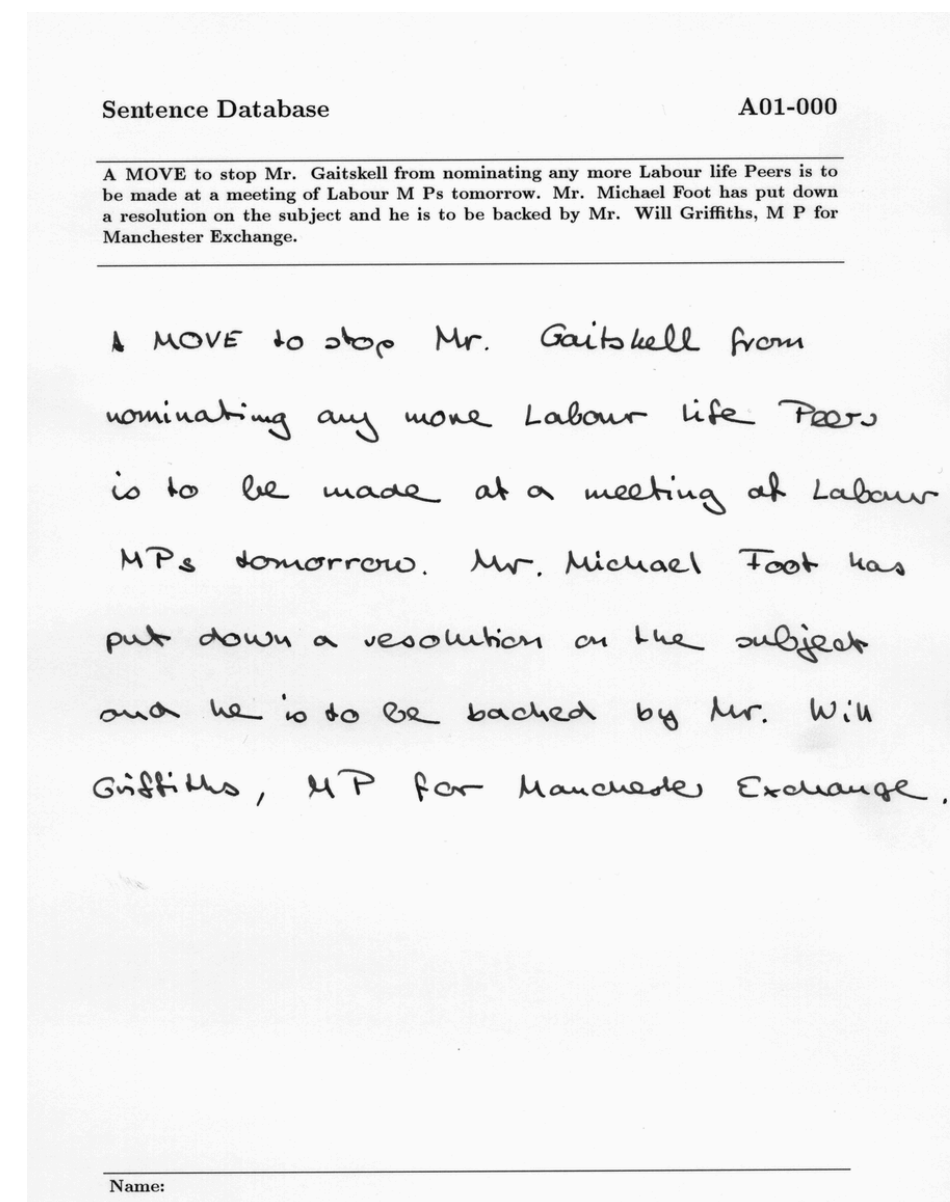


FIG 3: ORIGINAL DATASET

0208.png	The
0240.png	into
0235.png	to
0229.png	against
0213.png	saw to
0204.png	your
0231.png	a
0214.png	that.
0181.png	mice
0150.png	flat
0147.png	several
0188.png	That's
0146.png	take
0164.png	installation
0205.png	the
0189.png	rather
0158.png	running
0182.png	about
0185.png	soft

**FIG 4: ANNOTATION
FILE STRUCTURE**

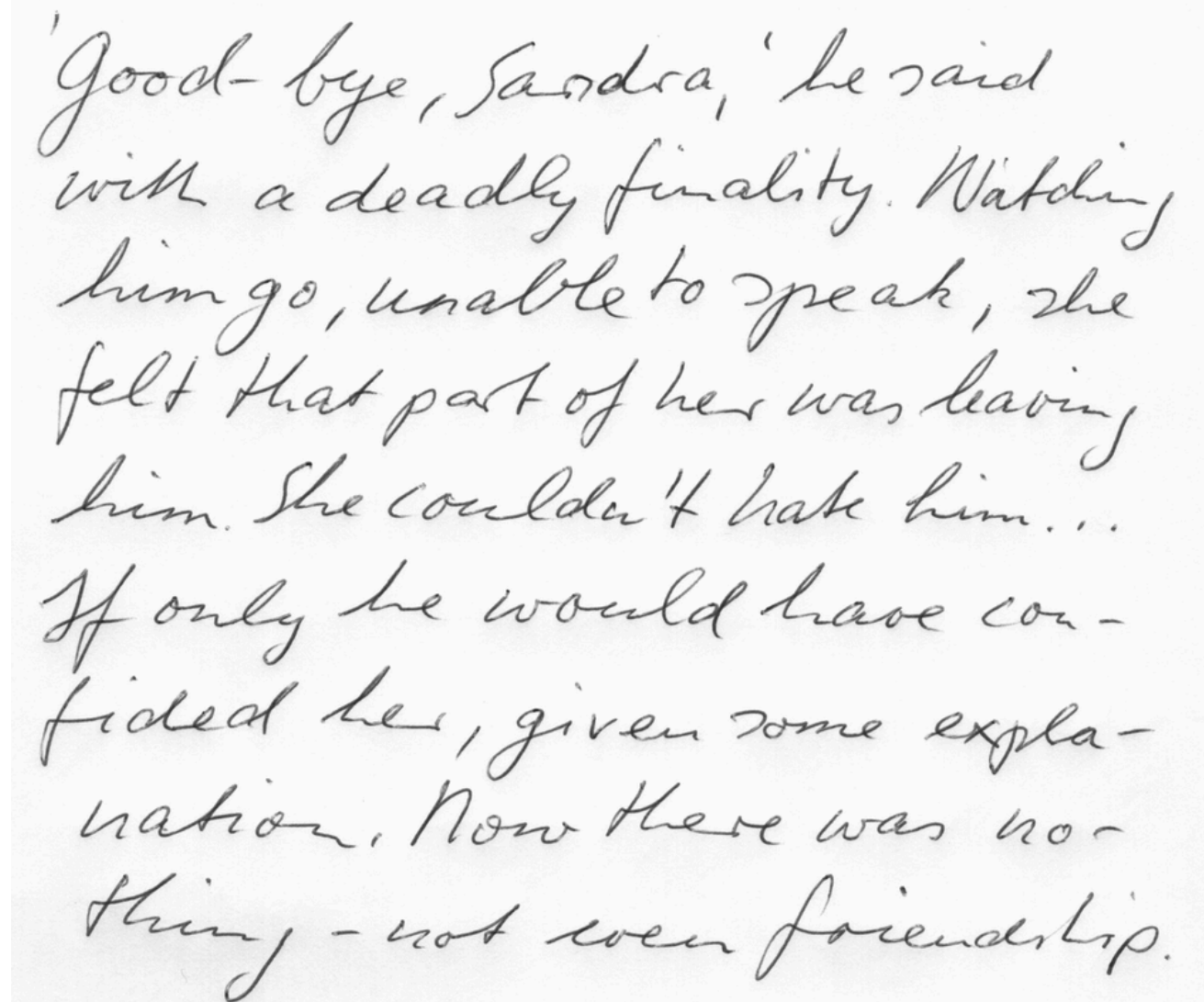
- Image Enhancements:
 - Noise reduction - Gaussian blur
 - Binarization and thresholding.
 - Skew correction -> misaligned text.
 - Morphological operations.
- Segmentation
 - Paragraph -> line segmentation.
 - Line -> word segmentation.

- PaddleOCR Engine:
 - Text detection model (e.g., DB, EAST).
 - Text recognition model (e.g., CRNN).
- TrOCR Model:
 - Vision Encoder Model - feature extraction
 - Text Decoder : BART
- Why use both TrOCR and PaddleOCR?

- Training:
 - Split annotated text labels -> tokens.
 - Convert tokens to a fixed length format.
 - Truncate
 - Padding
 - Assign ID's for tokens and decode
 - Eg: "hello world" → ["hello", "world"] → [874, 1231]
 - Apply custom training:
 - Custom trainer (Seq2SeqTrainer)
 - Cross Entropy Loss

- Validation:
 - Evaluation metrics:
 - Character Error Rate
 - Word Error Rate
 - Accuracy
 - Monitor overfitting/underfitting.
 - Test with validation data:
 - Adjust hyperparameters based on validation results.

- Error Correction:
 - Correct spelling mistakes.
 - Correct misrecognized words.
- Formatting : Fixing spaces, punctuation, capitalization etc.
- Contextual Analysis

A photograph of a handwritten text snippet on a light-colored background. The text is written in a cursive script and reads: 'Good-bye, Sandra,' he said with a deadly finality. Watching him go, unable to speak, she felt that part of her was leaving him. She couldn't hate him... If only he would have confided her, given some explanation. Now there was nothing - not even friendship.

'Good-bye, Sandra,' he said with a deadly finality. Watching him go, unable to speak, she felt that part of her was leaving him. She couldn't hate him... If only he would have confided her, given some explanation. Now there was nothing - not even friendship.

original

- Improves Visibility:
 - Enhance faint/unclear handwriting.
- Localized Contrast:
 - Adjust contrast regionally
 - Preserve details
 - Reduce noise.
- Boosts Accuracy:
 - Key preprocessing step for handling low-quality images.

- Set threshold value:
 - Distinguish different pixel intensities.
 - Classify foreground and background pixels.



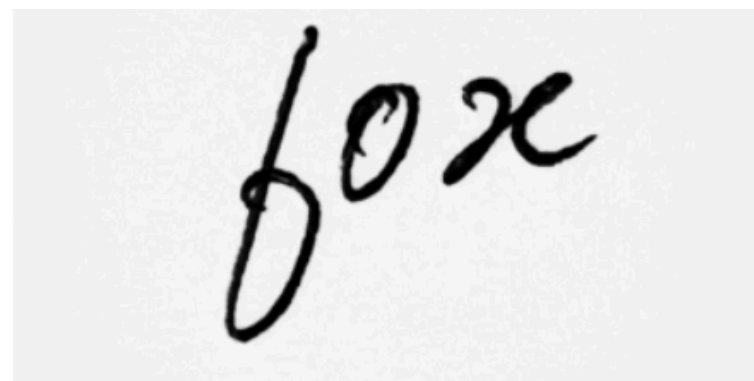
FIG 5: THRESHOLDING

- Otsu's method - distinct foreground and background.
- Niblack's method - variations in illumination/stroke thickness.
- Sauvola's method - noisy, degraded, or complex handwritten text.

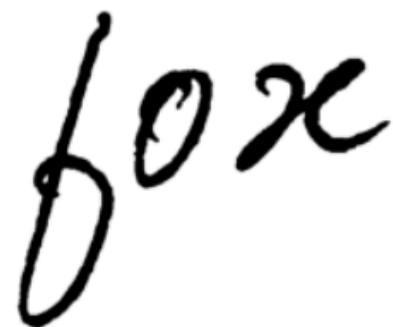


FIG 6: THRESHOLDING TECHNIQUES

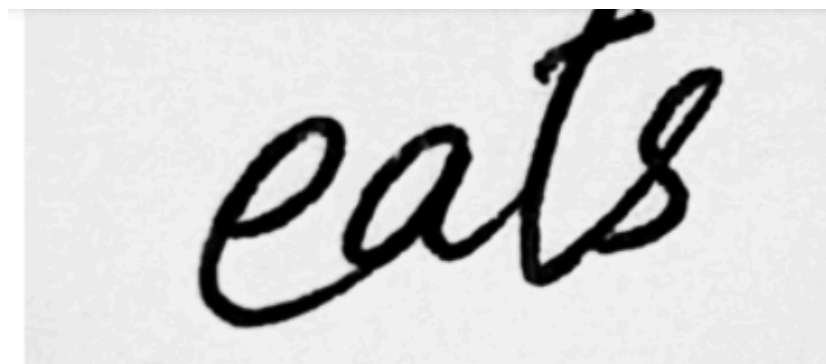
- Threshold hybrids - combine multiple thresholding methods.
- Otsu-Sauvola hybrid thresholding:
 - Splits image into regions.
 - Conditional rule - choose pixel/region-specific method
 - Optimal



Original Image: word_011.png



Otsu-Sauvola Hybrid Result

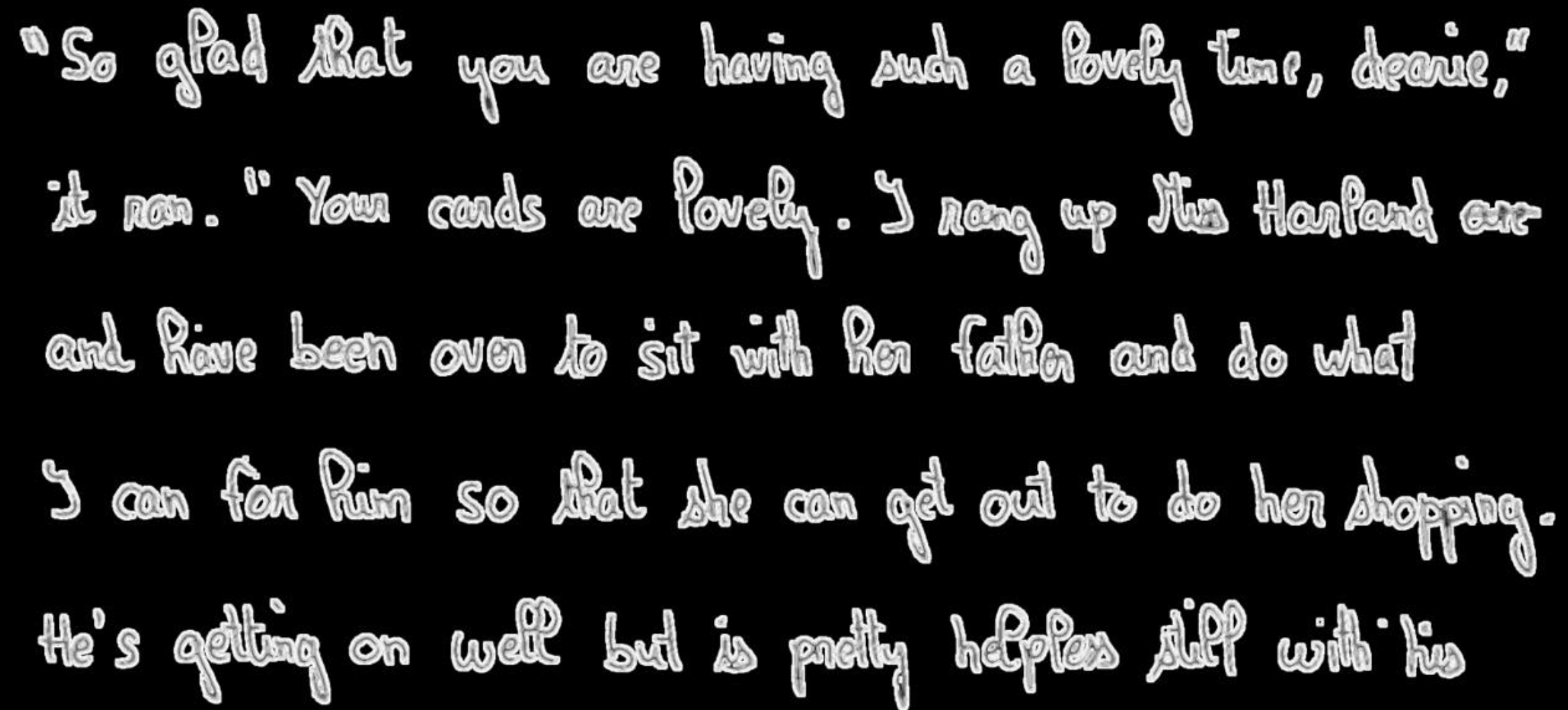


Original Image: word_010.png



Otsu-Sauvola Hybrid Result

FIG 7: OTSU-SAUVOLA HYBRID



"So glad that you are having such a lovely time, dearie,"
it ran. "Your cards are lovely. I rang up Miss Harland and
and have been over to sit with her father and do what
I can for him so that she can get out to do her shopping.
He's getting on well but is pretty helpless still with his

FIG 8: THRESHOLD OUTPUT

- Image Blending
 - Combine original and altered image
 - Masks/weights for smooth transitions.
 - Used in gradient-based merging.
- Unsharp Masking
 - Enhances sharpness
 - Adds edge details to blurred image.
 - Amplifies sharpness and clarity



'Good-bye, Sandra,' he said
with a deadly finality. Watching
him go, unable to speak, she
felt that part of her was leaving
him. She couldn't hate him...
If only he would have con-
fided her, given some expla-
nation. Now there was no-
thing - not even friendship.

original

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CLAHE

Otsu - Sauvola
thresholding, Image
Blending, Unsharp
Masking, Dilation

- Eliminate unwanted elements or distortions:
 - speckles
 - lines
 - irregular pixels.

There exist several methods to design forms with fields to fields may be surrounded by bounding boxes, by light rectangles or methods specify where to write and, therefore, minimize the effect with other parts of the form. These guides can be located on a sheet is located below the form or they can be printed directly on the form a separate sheet is much better from the point of view of the quality but requires giving more instructions and, more importantly, rest this type of acquisition is used. Guiding rulers printed on the used for this reason. Light rectangles can be removed more easily whenever the handwritten text touches the rulers. Nevertheless, be taken into account: The best way to print these light rectangles

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FIG 9: NOISE REMOVAL

Denoised Image 4

'Good-bye, Sandra,' he said with a deadly finality. Watching him go, unable to speak, she felt that part of her was leaving him. She couldn't hate him... If only he would have confided her, given some explanation. Now there was nothing - not even friendship.

Histogram of Denoised Image 4

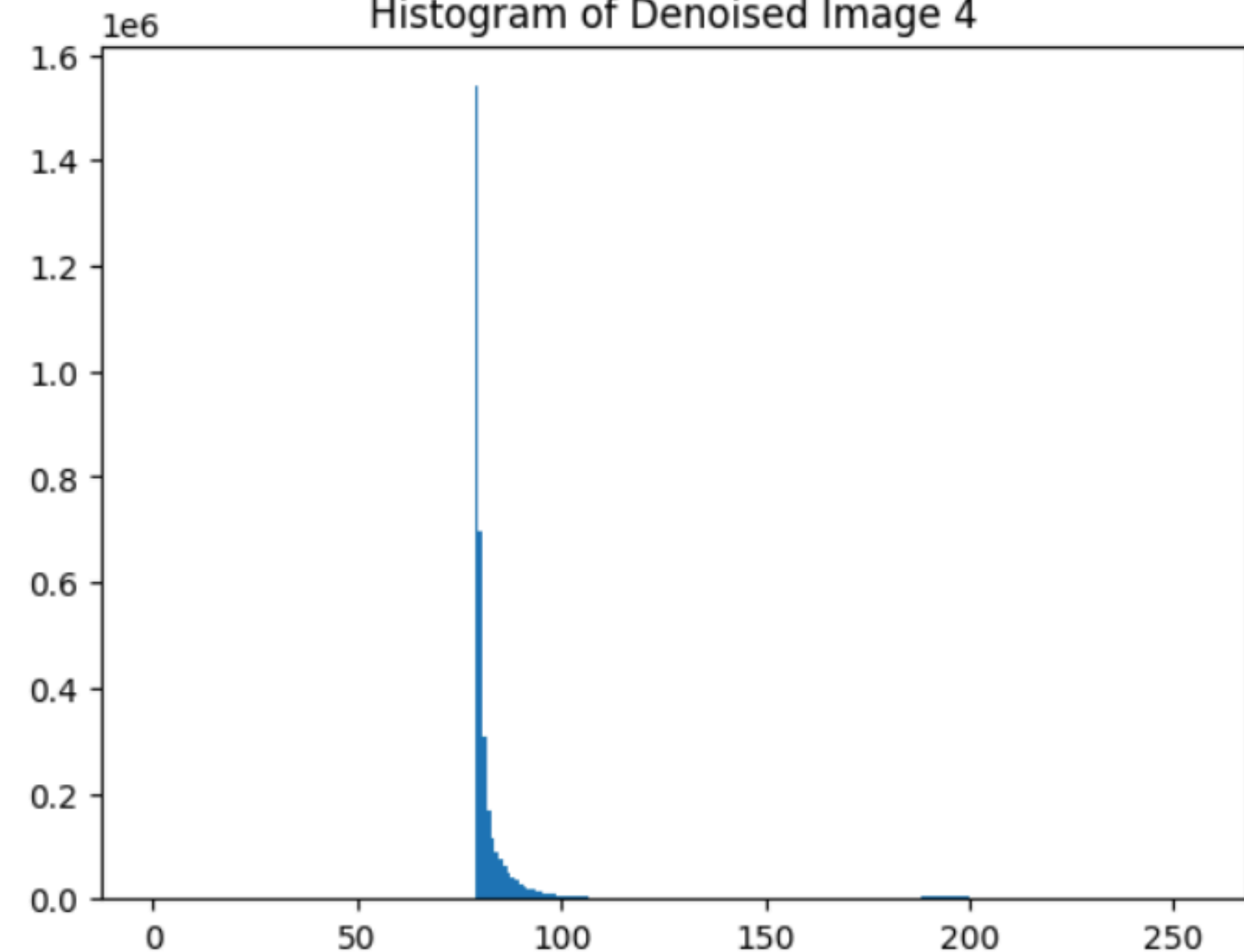
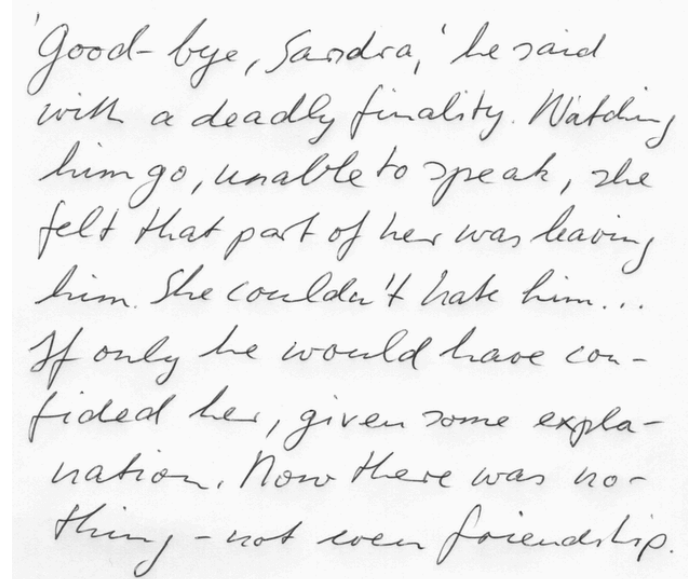
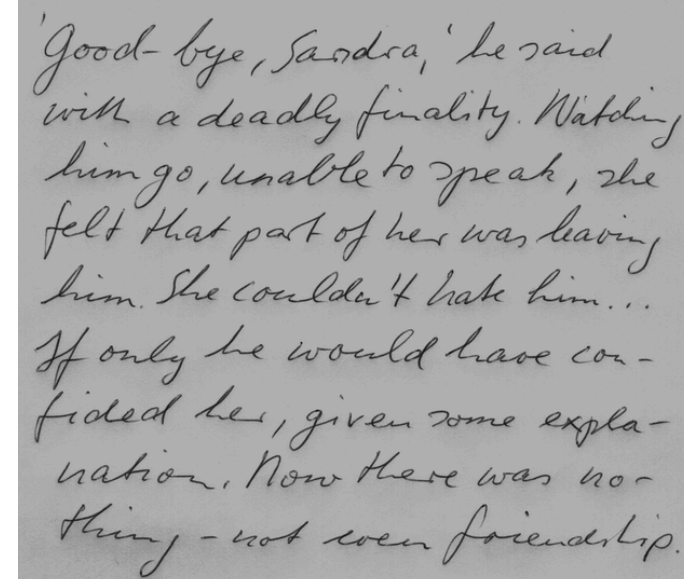


FIG 10: DENOISED OUTPUT



'Good-bye, Sandra,' he said
with a deadly finality. Watching
him go, unable to speak, she
felt that part of her was leaving
him. She couldn't hate him...
If only he would have con-
fided her, given some expla-
nation. Now there was no-
thing - not even friendship.

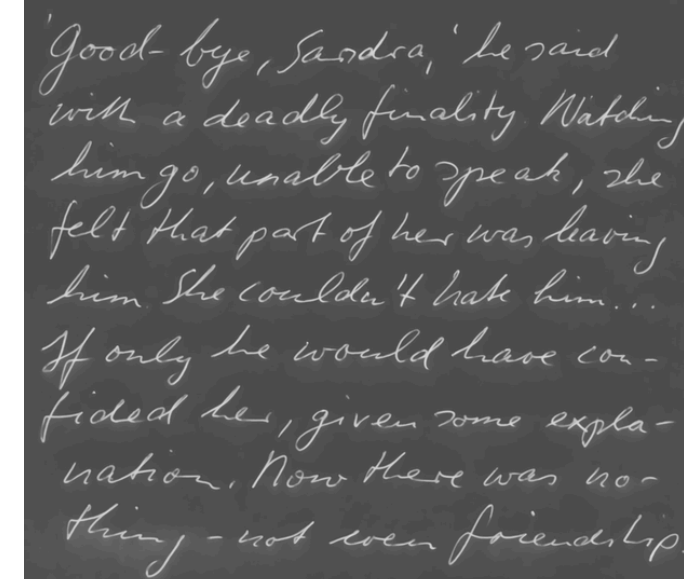
original



'Good-bye, Sandra,' he said
with a deadly finality. Watching
him go, unable to speak, she
felt that part of her was leaving
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CLAHE

Otsu - Sauvola
thresholding, Image
Blending, Unsharp
Masking, Dilation



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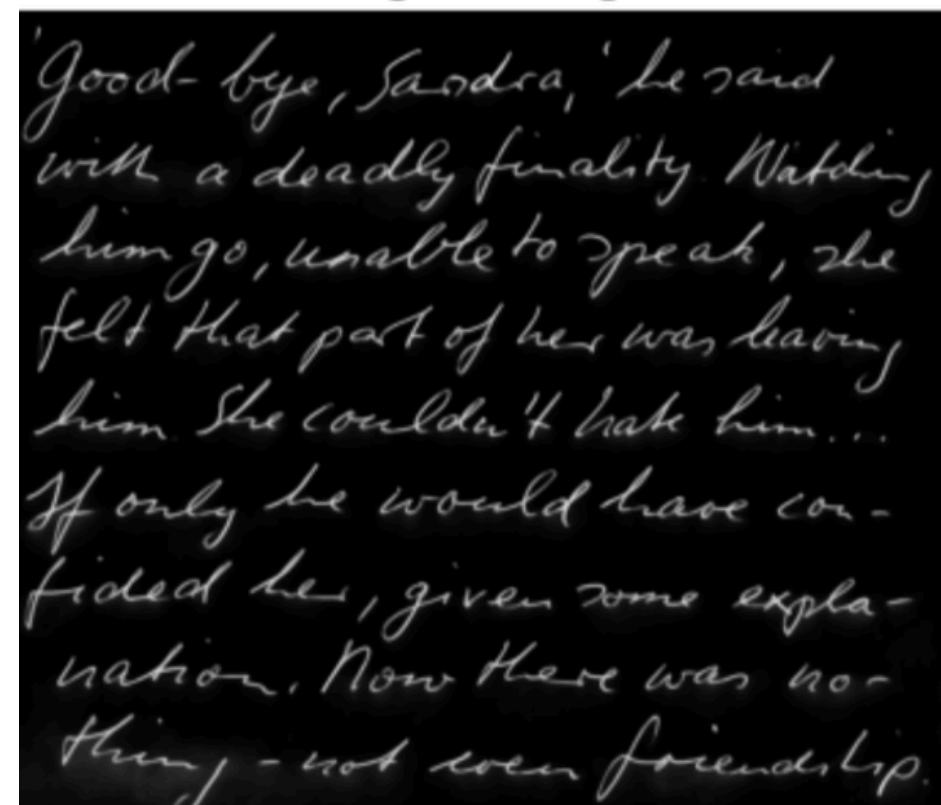
Denoising

- Simplify the image - black and white
- Better text recognition.



FIG 11: BINARIZATION

Original Image



Binarized Image

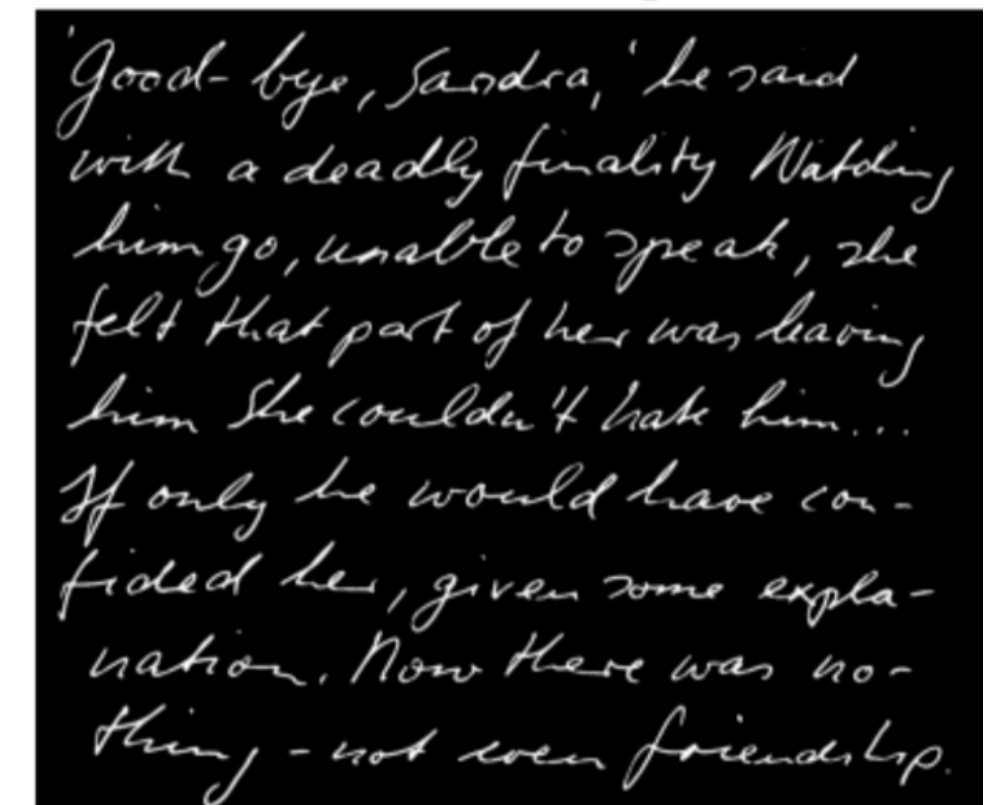
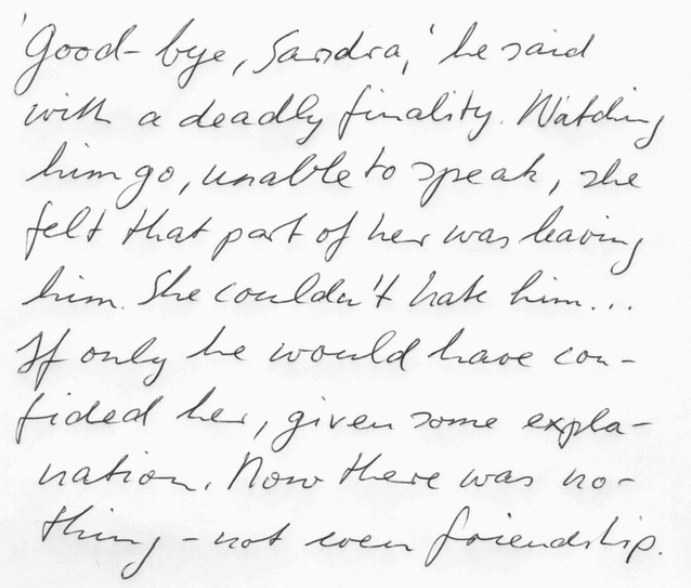


FIG 12: BINARIZED OUTPUT

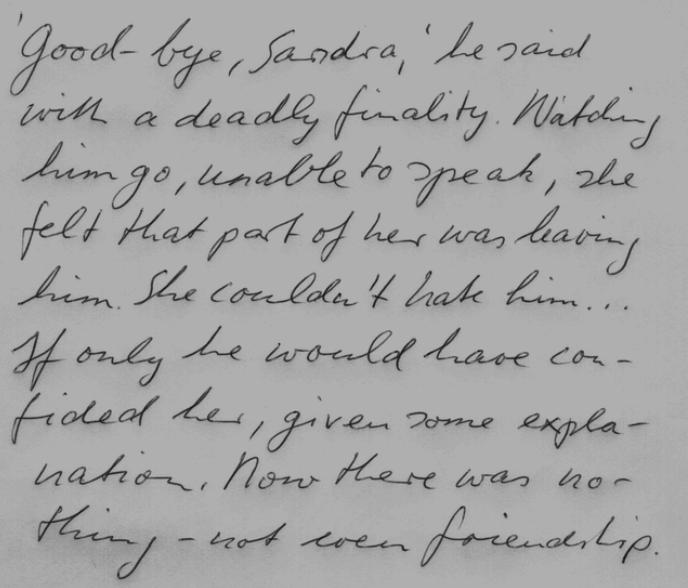
PREPROCESSING

WORK DONE



'Good-bye, Sandra,' he said
with a deadly finality. Watching
him go, unable to speak, she
felt that part of her was leaving
him. She couldn't hate him...
If only he would have con-
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nation. Now there was no-
thing - not even friendship.

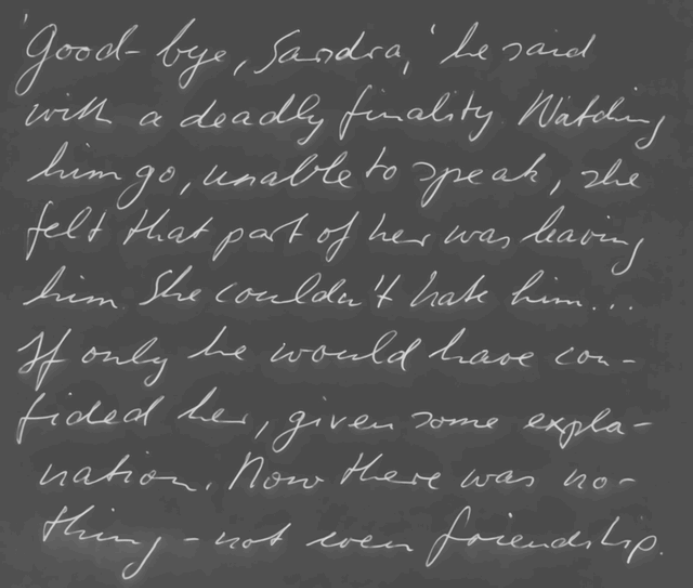
original



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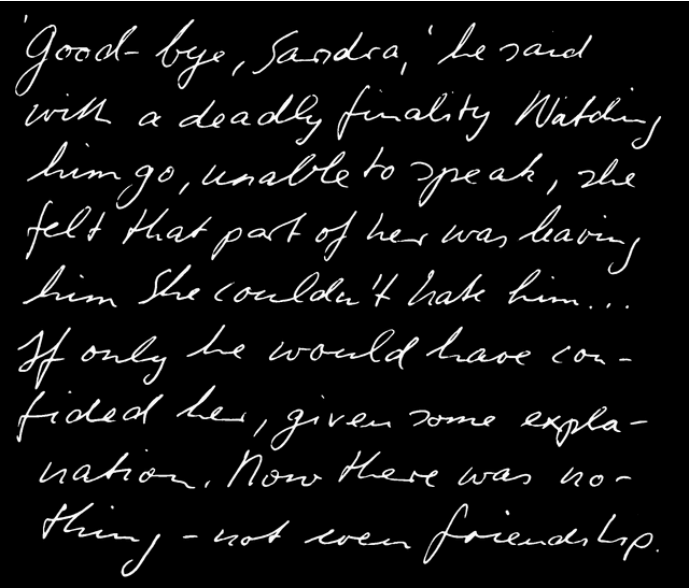
CLAHE

Otsu - Sauvola
thresholding, Image
Blending, Unsharp
Masking, Dilation



'Good-bye, Sandra,' he said
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Denoising



'Good-bye, Sandra,' he said
with a deadly finality. Watching
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felt that part of her was leaving
him. She couldn't hate him...
If only he would have con-
fided her, given some expla-
nation. Now there was no-
thing - not even friendship.

Binarization

- Correct orientation of tilted/slanted images:
 - `correct_skew()`
 - Projection Profile
 - Hough and Affine Transformation

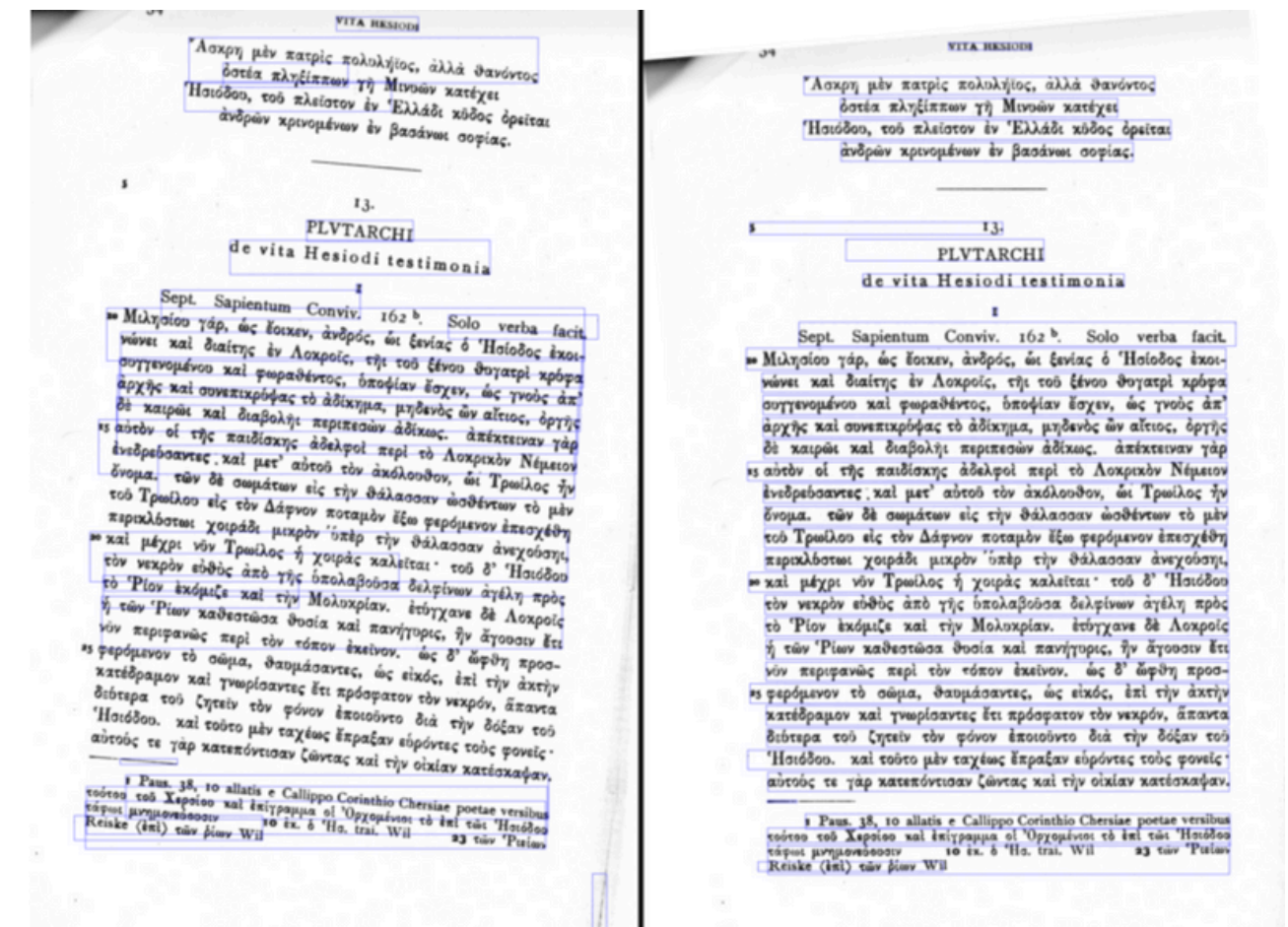


FIG 13: DESKEWING

- Canny edge detection
- Feature extraction technique
 - Detect lines in the hough space
- Deskews images



FIG 14: HOUGH TRANSFORM

Dilation

- Expands foreground object boundaries.

Erosion

- Shrinks foreground object boundaries.



FIG 15: DILATION & EROSION

Morphological Closing:

- Dilation operation, then erosion.
- Fill small gaps in text or objects -> improves OCR accuracy.

Contour Detection:

- Trace object boundaries
 - Intensity
 - Color differences.
- Measure area, perimeter, and bounding boxes of objects.

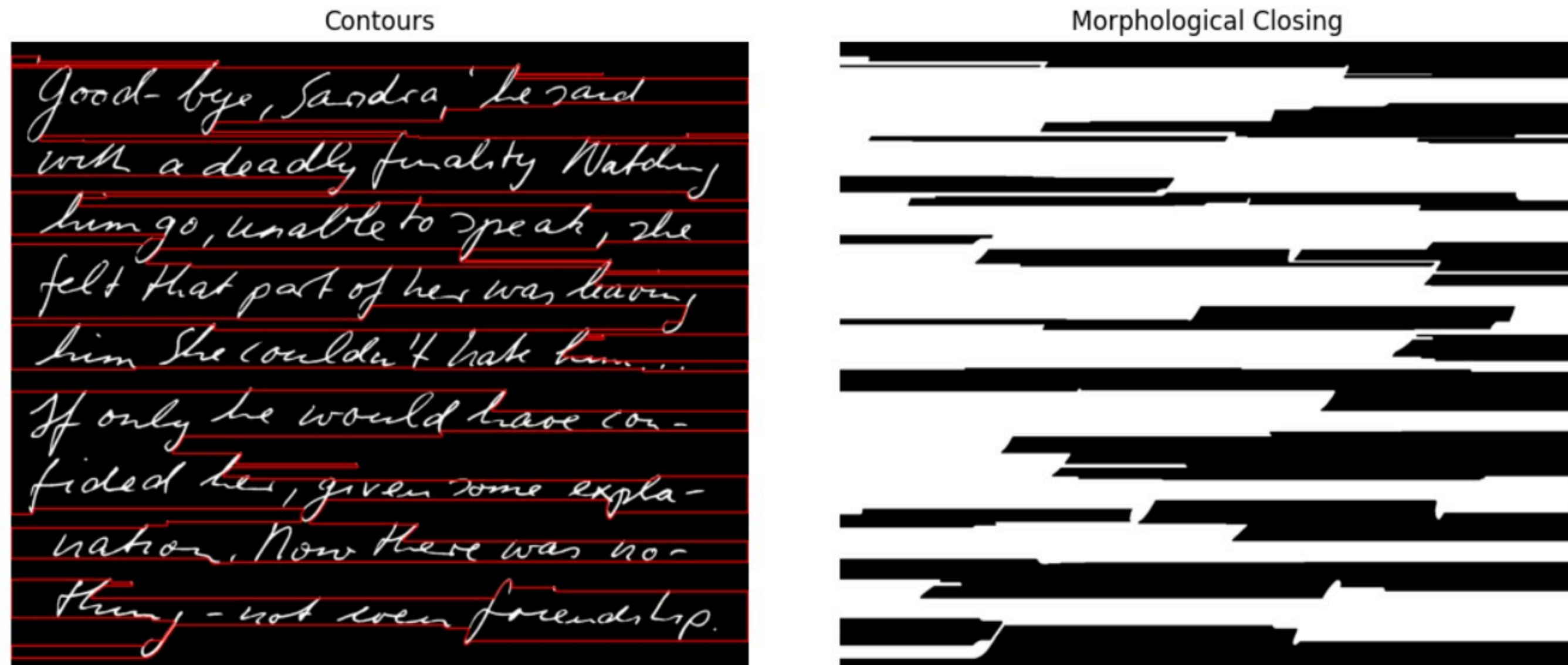
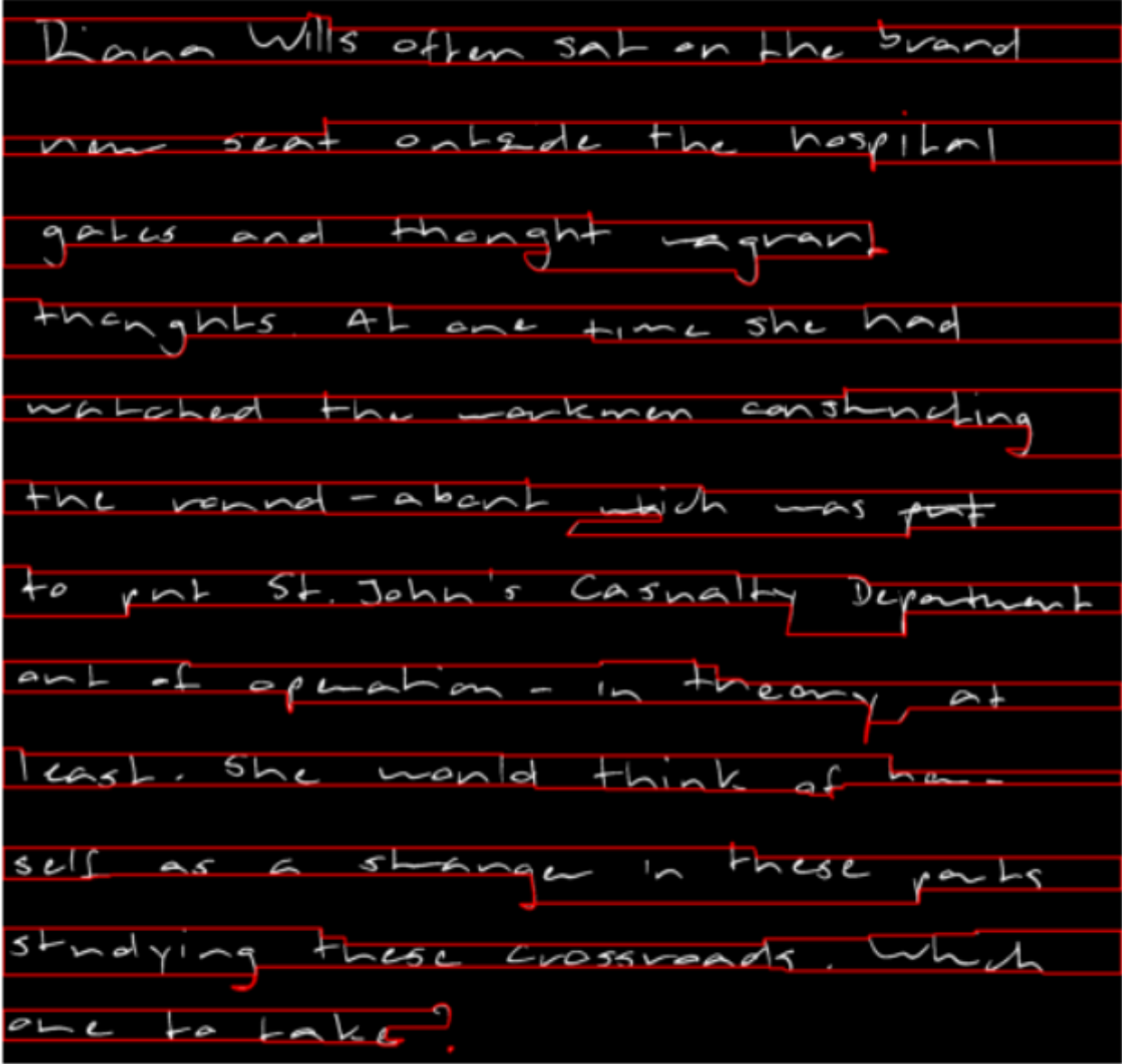


FIG 16: MORPHOLOGICAL OPERATION

Contours

A black rectangular area containing handwritten text in white. Red lines are drawn over the text, highlighting specific words and phrases. The text is: "Diana Willis often sat on the brand new seat outside the hospital gates and thought vagrant thoughts. At one time she had watched the workmen constructing the round - about which was put to put St. John's Casualty Department out of operation - in theory, at least. She would think of herself as a stranger in these parts studying these crossroads. Which one to take?"

Diana Willis often sat on the brand new seat outside the hospital gates and thought vagrant thoughts. At one time she had watched the workmen constructing the round - about which was put to put St. John's Casualty Department out of operation - in theory, at least. She would think of herself as a stranger in these parts studying these crossroads. Which one to take?

Morphological Closing

A black rectangular area containing handwritten text in white. The text is completely obscured by thick black horizontal bars, representing the result of a morphological closing operation.

[REDACTED]

FIG 17: MORPHOLOGICAL OPERATION

Line Segmentation:

- Morphological closing - fill gaps between characters
- Contour detection - identify bounding boxes per line.

Word Segmentation:

- Morphological closing - character connection
- Contour detection - isolate individual words.

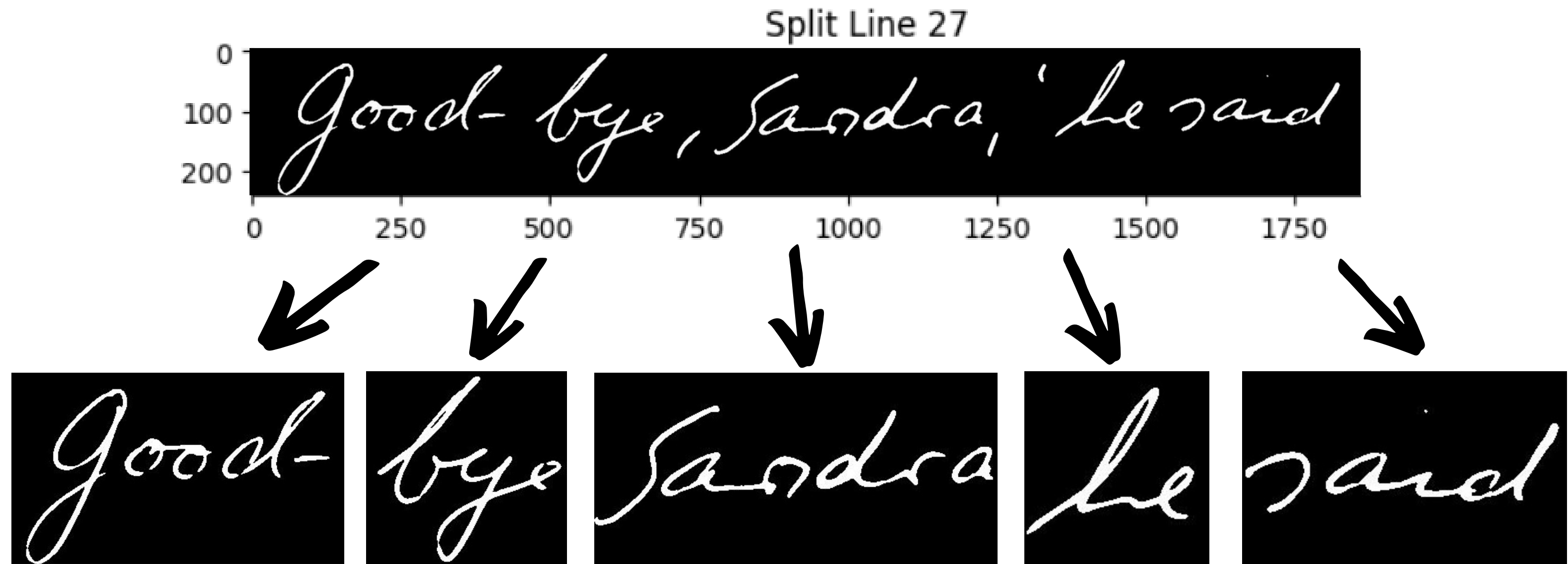


FIG 18: SEGMENTATION

- Load Model and Processor
- Create Custom Dataset Class
- Model Configuraion
- Define Training Arguments and create Custom Trainer
- Execute `Trainer.train()`

- Default search method in TrOCR - greedy search
 - num_beams = 1
- Beam search used in text generation tasks:
 - decoding algorithm
 - num_beams = 4
 - Keeps four possible sequences.

- Prevent overfitting/underfitting:
 - Modify learning rate, batch size, and other parameters.
- Two Approaches:
 - Validation-Based Tuning → Evaluate CER, WER, and Accuracy
 - Logging & Tracking → Store training/evaluation loss
 - Analyze model behavior.

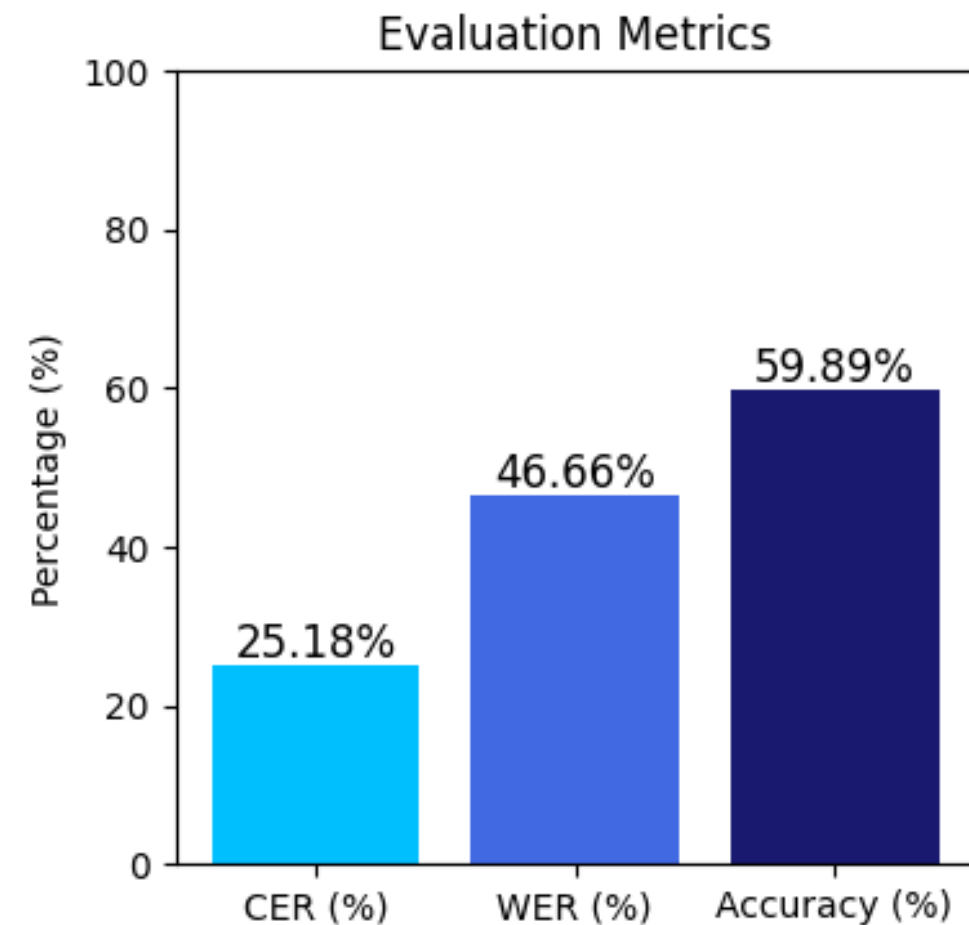
- Learning Rate
- Batch Size
- Number of Epochs
- Beam Search
- Weight Decay

EVALUATION

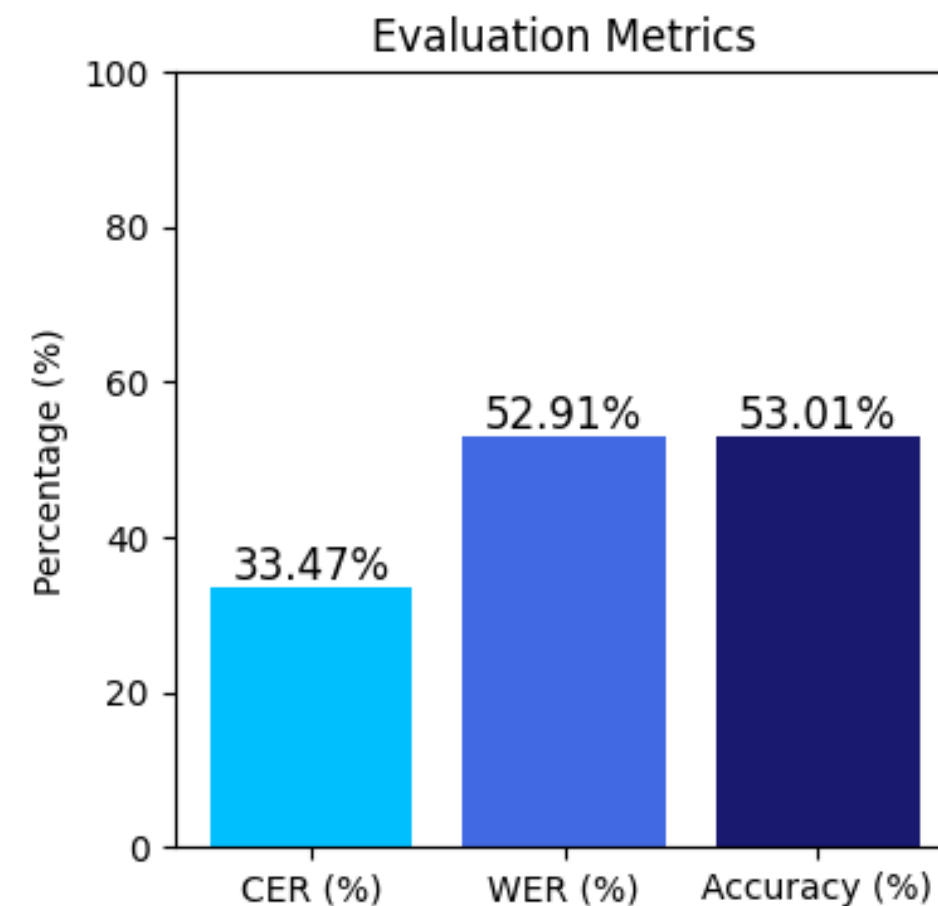
WORK DONE

	IDEAL	GOOD	ACCEPTABLE	HIGH
CER	0%	1-5%	5-10%	>10%
WER	0%	5-10%	10-20%	>20%

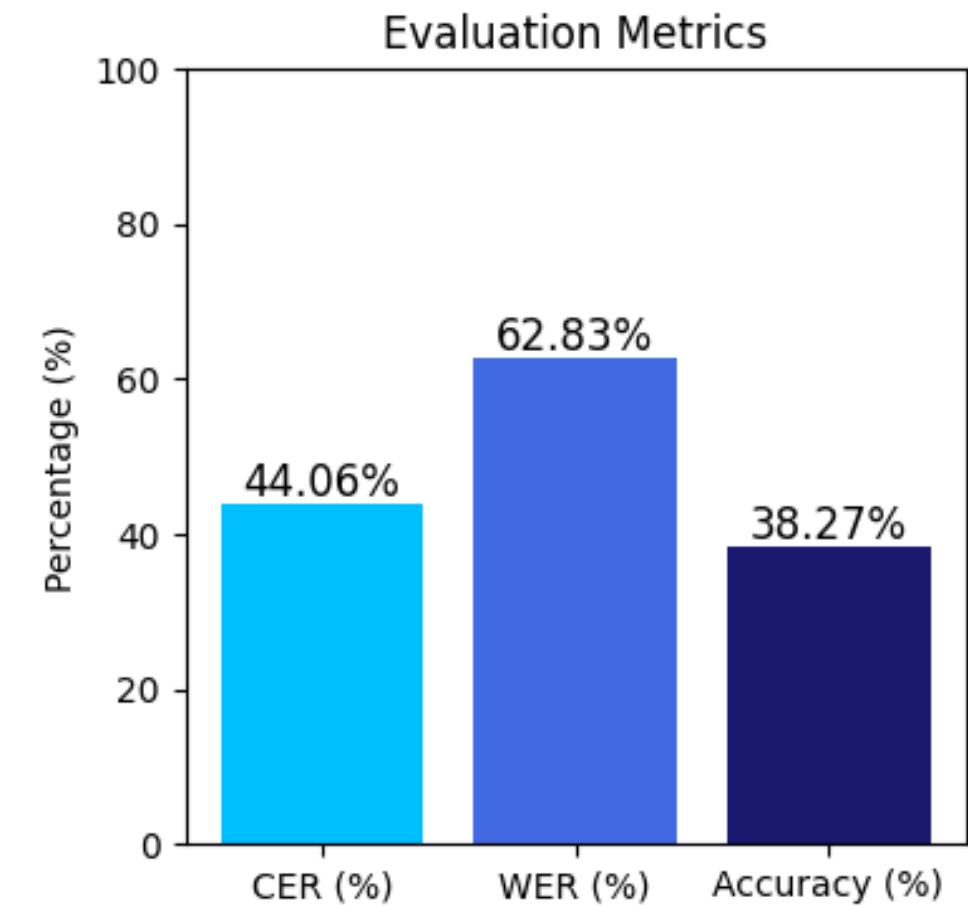
FIG 19: EVALUATION METRIC



Train-loss : 1.7543
Eval-loss: 1.3515

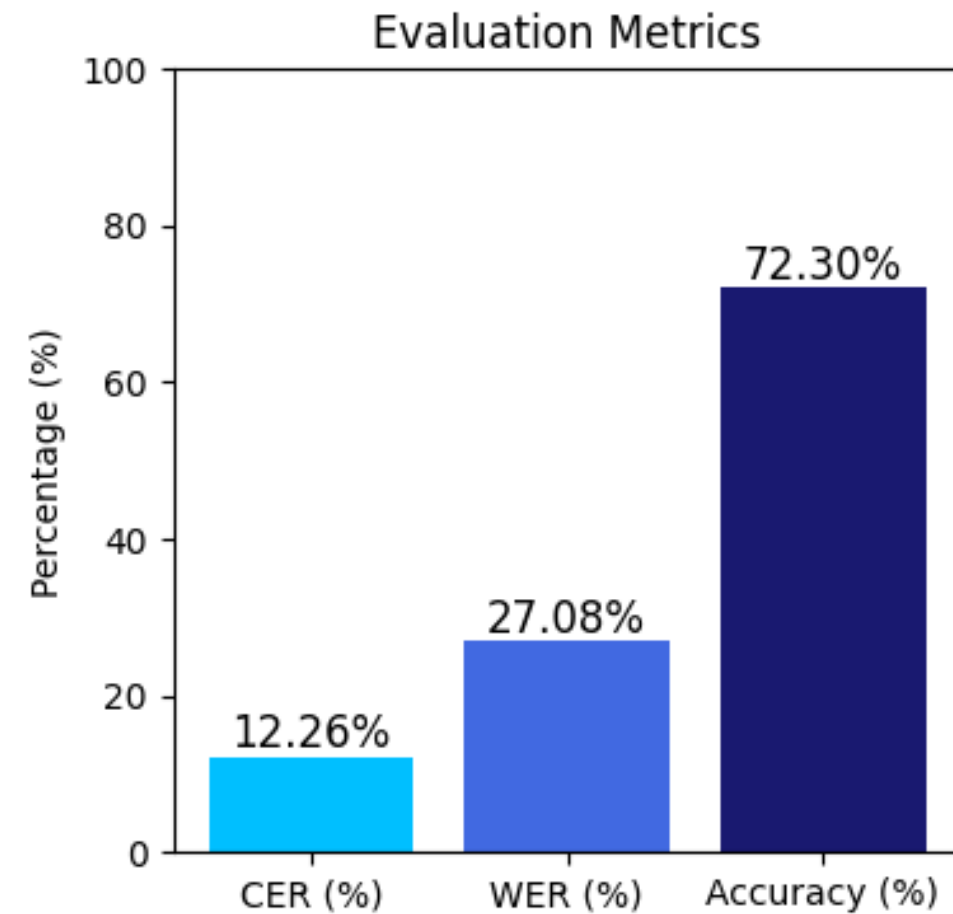


Train-loss : 0.8632
Eval-loss: 1.6241

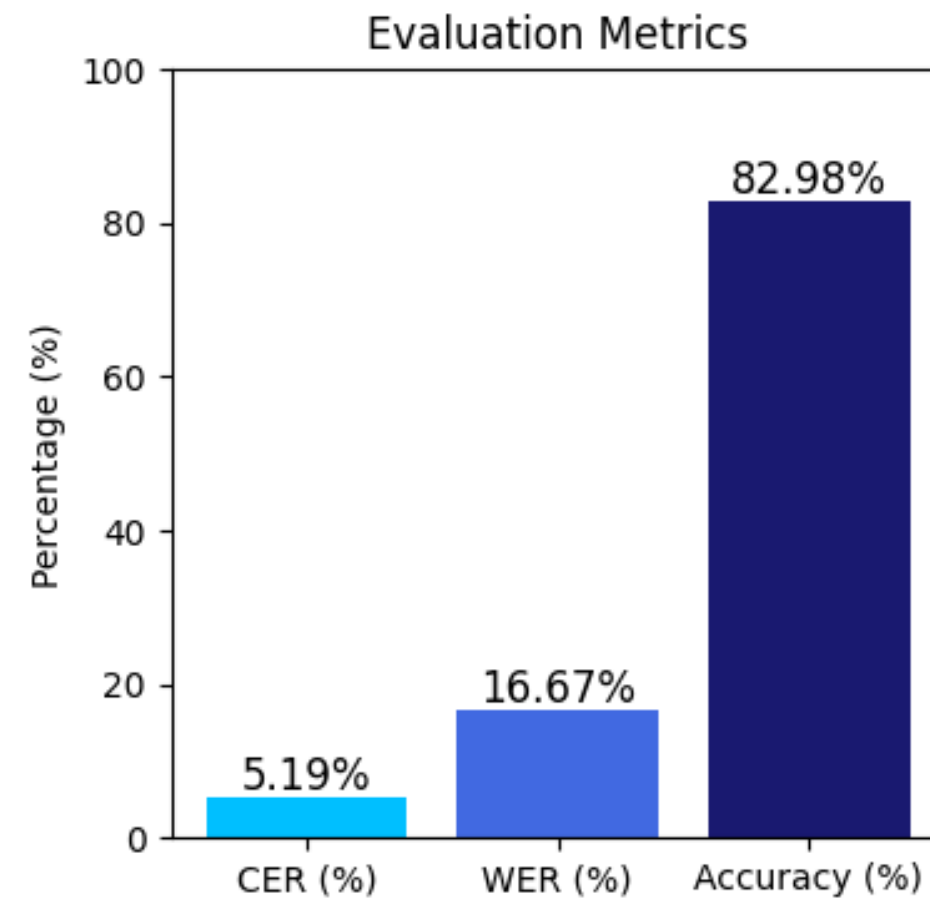


Train-loss : 0.4263
Eval-loss: 1.7517

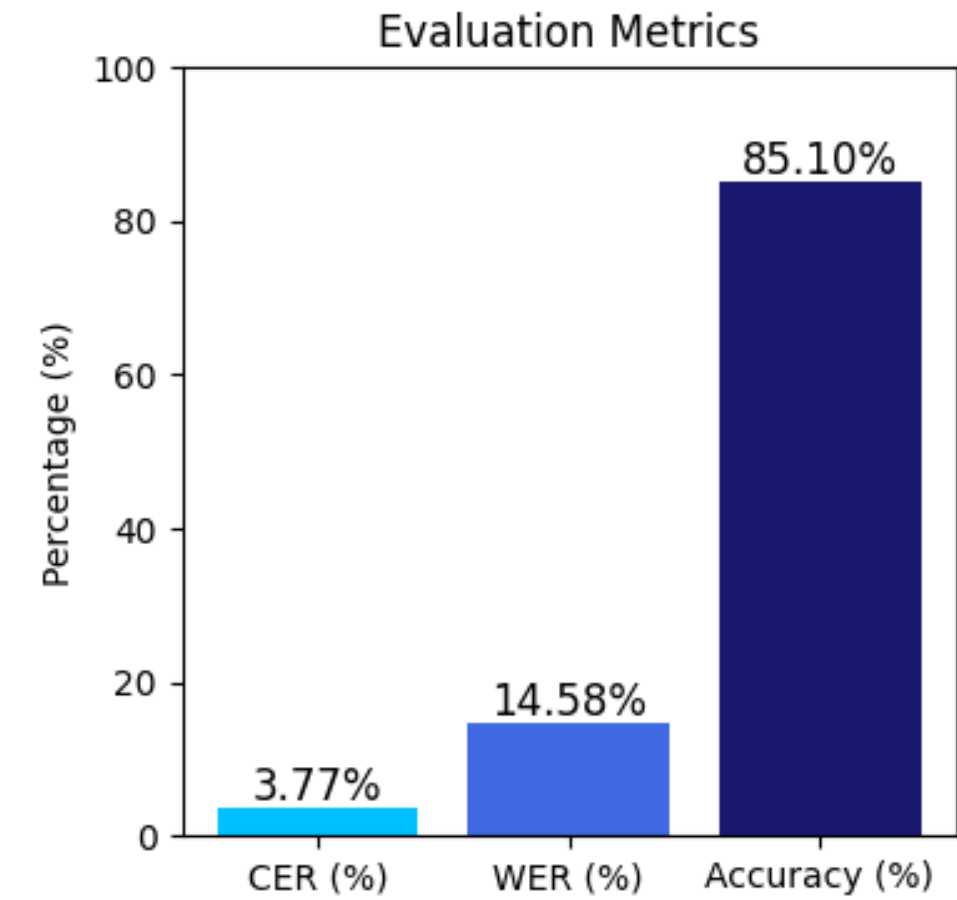
FIG 20: INITIAL TRAINING RESULTS



Train-loss : 6.4060
Eval-loss: 1.4407



Train-loss : 2.0652
Eval-loss: 0.9568



Train-loss : 2.0052
Eval-loss: 0.9268

FIG 21: FINAL TRAINING RESULTS

Predicted: inspector.



Predicted: outside



Predicted: and agreed



FIG 22: VALIDATION RESULTS

Predicted: nominating any more Labour life



Recognized Text: nominating any more Labour life

Predicted: and he is to be backed by Mr. will



Recognized Text: and he is to be backed by Mr. will

Predicted: I Move to stop Mr. Gaiblell from



Recognized Text: I Move to stop Mr. Gaiblell from

FIG 23: FINE-TUNED TROCR RESULTS FOR
SINGLE-LINE TEXT

- Multiline/Paragraph recognition.
- Hindered processing accuracy in full handwritten documents.
 - Solution: Hybrid OCR Approach
 - PaddleOCR:
 - Detects and recognizes text regions
 - Output: segmented handwritten text
 - TrOCR:
 - Enhanced recognition accuracy.

A MOVE to stop Mr. Gaitskell from nominating any more Labour life Peers is to be made at a meeting of Labour MPs tomorrow. Mr. Michael Foot has put down a resolution on the subject and he is to be backed by Mr. Will Griffiths, MP for Manchester Exchange.

PaddleOCR Metrics:

Accuracy: 35.55%

CER: 0.64

WER: 0.92

TrOCR Metrics:

Accuracy: 0.39%

CER: 1.00

WER: 1.00

Hybrid (PaddleOCR + TrOCR) Metrics:

Accuracy: 51.95%

CER: 0.48

WER: 0.67

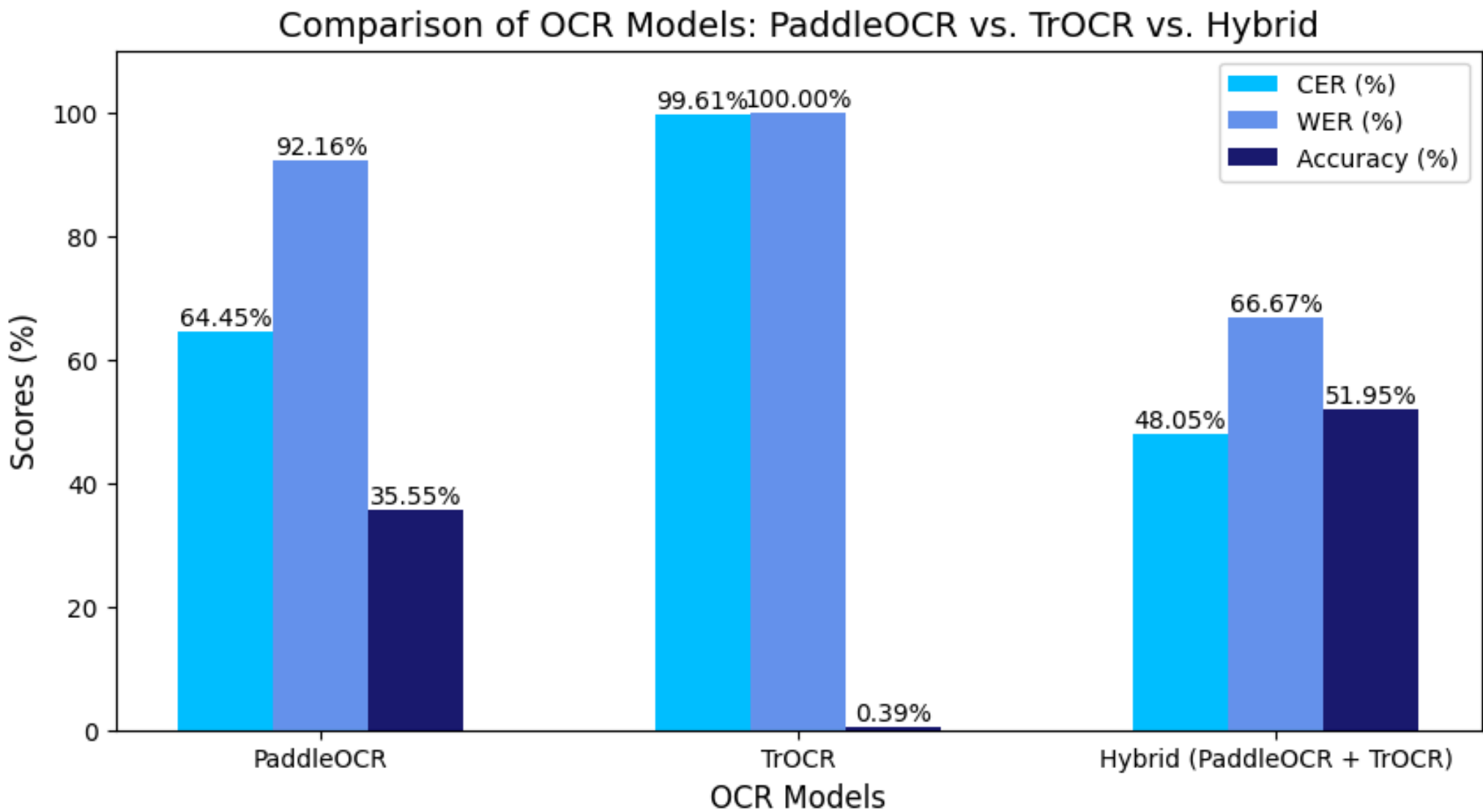


FIG 24: OCR MODEL RESULTS FOR HANDWRITEN MULTI-LINE TEXT

HYBRID OCR OUTPUT

WORK DONE

Though they may gather some left-wing support, a large majority of Labour MPs are likely to turn down the Foot-Griffiths resolution. Mr. Foot's line will be that as Labour MPs opposed the Government Bill which brought life peers into existence, they should not now put

=== Final Recognized Text ===

Though they may may gather some left-wing support, a large majority at labour Mrs are likely to turn down the foot-Griffitus resolution. Mr.will will be that as Labour MPs opposed Government Bil which brought life means into evidence, theynot not now put

Processing Line 1...

Though they may gather some left-wing

Processing Line 2...

support, a large majority of Labour

Processing Line 3...

MPs are likely to turn down the Foot-

Processing Line 4...

Griffiths resolution. Mr. Foot's line will

Processing Line 5...

be that as Labour MPs opposed the

Processing Line 6...

Government Bill which brought life peers

Processing Line 7...

into existence, they should not now put

=== Final Recognized Text ===

Though they may may gather some left-wing support, a large majority at labour Mrs are likely to turn down the foot-Griffitus resolution. Mr.will will be that as Labour MPs opposed Government Bil which brought life means into evidence, theynot not now put

FIG 25: HYBRID OCR RESULTS

1. Text Normalization
2. Spell Correction
3. Punctuation Restoration
4. Sentence Segmentation
5. Context-Aware Correction

- Purpose: Removes noise/redundancy from raw OCR output.
- How:
 - Replaces multiple whitespaces -> single space.
 - Removes duplicate words (e.g., "the the" → "the").
 - Filters out non-alphanumeric symbols.

- Purpose: Fixes OCR-induced spelling errors.
- How:
 - Uses pyspellchecker to find and correct misspelled words.
 - Custom whitelist preserves domain-specific acronyms (e.g., AI, OCR).

- Purpose: Adds missing punctuation marks.
- How:
 - Uses a pre-trained transformer model
 - ‘fullstop-punctuation-multilang-large’
 - Infers punctuation based on context.

- Purpose: Fixes grammar and awkward phrasing that may pass through previous steps.
- How:
 - Uses a fine-tuned grammar correction model
 - flan-t5-base-grammar-synthesis
 - Operates on full paragraphs -> better context.

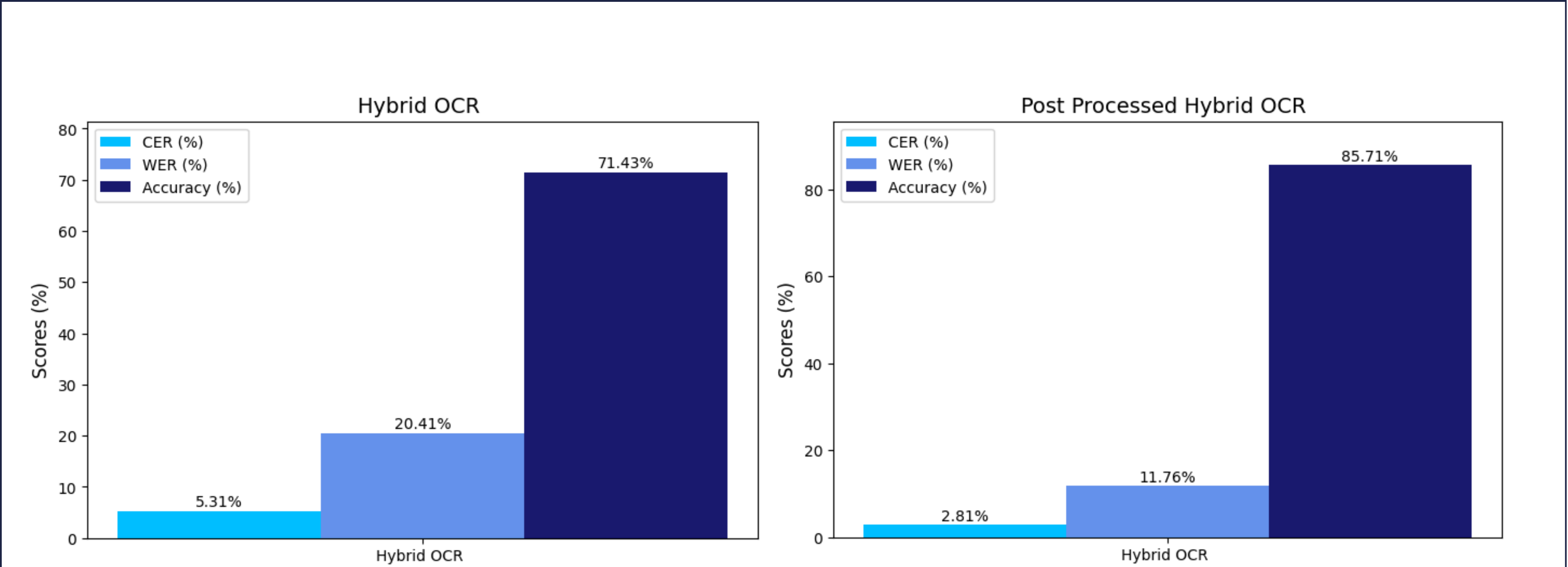


FIG 26: BEFORE POST-PROCESSING

FIG 27: AFTER POST-PROCESSING

=== Final Recognized Text ===
Though they may may gather some left-wing support, a large majority at labour Mrs are likely to turn down the foot-Griffitus resolution. Mr.will will be that as Labour MPs opposed Government Bil which brought life means into evidence, theynot not now put

FIG 28: BEFORE POST-PROCESSING

=== Final Recognized Text ===
Though they may gather some Left-wing support , a large majority of Labour . MPs are likely to turn down the Foot - Griffiths resolution . Mr. Foot's line will be that as Labour MPs opposed the Government Bill which brought life peers into existence , they should not now put

FIG 29: AFTER POST-PROCESSING

=== Line 1 ===
Though they may gather some Left-wing
=== Line 2 ===
support, a large majority of Labour
=== Line 3 ===
MPs are likely to turn down the Foot -
=== Line 4 ===
Griffiths resolution. Mr. Foot's line will
=== Line 5 ===
be that as Labour MPs opposed the
=== Line 6 ===
Government Bill which brought life peers
=== Line 7 ===
into existence, they should not now put
=== Final Recognized Text ===
Though they may gather some Left-wing support , a large majority of Labour . MPs are likely to turn down the Foot - Griffiths resolution . Mr. Foot's line will be that as Labour MPs opposed the Government Bill which brought life peers into existence , they should not now put

FIG 30: FINAL RESULTS

TIMELINE

MONTH		SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER	JANUARY	FEBRUARY	MARCH
Project Planning								
Literature Review								
Data Collection								
Architecture Design								
IMPLEMENTATION	Model Selection/Finetuning							
	Preprocessing							
	OCR Enhancements Intensity							
	Web Application							
Testing & Performance Evaluation								
Project Delivery								
Review Paper								
Research Paper Publish								

CONCLUSION

- Hybrid OCR System:
 - PaddleOCR - detection
 - TrOCR - enhanced recognition.
- Improved Multiline Recognition
- Enhanced Accuracy
 - Advanced preprocessing
 - Fine-tuned models.
- Future Scope:
 - Transformer-based models for post-processing
 - Real-time deployment.

PROGRAM OUTCOMES

PO Number	Program Outcome	Project Application
PO 1	Engineering Knowledge	Built OCR using PaddleOCR and TrOCR.
PO 2	Problem Analysis	Solved handwritten text recognition and scalability issues.
PO 3	Design & Development	Designed hybrid OCR with preprocessing.
PO 4	Investigate Complex Problems	Handled cursive and multi-line inputs.
PO 5	Modern Tool Usage	Used PaddleOCR (CRNN) and TrOCR (Transformer).
PO 6	The Engineer and Society	Improved accessibility via digitization.

PROGRAM OUTCOMES

PO Number	Program Outcome	Project Application
PO 7	Environment and Sustainability	Reduced paper use through OCR.
PO 8	Ethics	Maintained data privacy.
PO 9	Individual and Team Work	Collaborated by dividing OCR pipeline into modules.
PO 10	Communication	Shared technical progress and system design.
PO 11	Project Management	Planned and executed development phases.
PO 12	Lifelong Learning	Explored evolving OCR models and tools.

CONTRIBUTION

Phases	Team Member
Data Collection	Hisham & Saraswathy
Preprocessing	Hisham
Recognition	Chaitanya
Training & Validation	Chaitanya
Postprocessing	Saraswathy

THANK YOU!